IMP - Internal Monitoring Plan

Plan identification, purpose and boundary

SIMULATED · v1.0

Study / protocol GLENOS-01 / GS-2026-01	**CIP version** CIP v4.2	**Site / code** Riverside Orthopedic Center / 014	**IMP ID** IMP-014
Version v1.1	**Status** ☐ Draft ☑ Approved	**Supersedes (ID / version)** IMP-014 v1.0 (approved 12 Aug 2026)	**Effective from** 4 Nov 2026
Next scheduled plan review 4 May 2027	**Plan owner / role** Maria Silva / Study Coordinator	**Approved by / date** Dr Alex Morgan / 3 Nov 2026	**Controlled location** Investigator site file, tab 3 (Quality) / IMP-014

A Purpose and use

The IMP is the site's plan for checking its own work during the investigation, alongside the sponsor's monitoring. It is written and approved before the first participant is included. It does not record any review: each review is run and recorded in the Internal Monitoring Checklist (IMC), which cites the IMP version it followed.

For each of the twelve checks, and any check the site adds, the plan fixes in advance what is checked (C), when, by whom, on which records and how many (D), and what counts as done, evidenced and effective (E). It also sets what calls a review early (F) and where each kind of result goes (G). Rules are set before results exist. A change applies only from its effective date onward and is entered in I; reviews already done stay with the version they used.

Unit of analysis	Decision supported	Inputs	Outputs
One site and one investigation, under one approved version of the plan.	Whether the site's own verification is defined well enough to catch, in time, a failure in a control the site is responsible for.	Site risk register (SRR), delegation log and training records, the CIP and its manuals, and the records each check reads.	The rules every IMC review follows. Findings go to the IMC, corrective actions to the CAPA log, plan changes back to I.

B Plan boundary

What the plan covers and what it rests on. The plan covers controls the site is responsible for; it does not replace sponsor monitoring.

Investigation(s), sites and phases covered	GLENOS-01 at site 014 only; from start-up to close-out.
Documents the plan is built on (ID / version)	CIP v4.2; imaging manual v2.0; SRR-014 v3; delegation log rev. 7; source document map v1.2.
Sponsor monitoring: what the sponsor checks and how its findings reach the site	Sponsor CRA visit every 8 weeks with 20% source check of endpoint fields. Findings arrive by follow-up letter to the PI and the coordinator.
Known constraints (staff, time, access to records)	One coordinator does most data entry. Radiology archive open to the team on weekdays only. PI on leave 27 Oct to 2 Nov 2026.
If a record picked for review is unavailable (rule for every check)	Logged in the IMC as unavailable; the next unit in the list is taken. If more than 1 in 5 picked units are unavailable, the check is rated not assessable.
Outside this plan	Sponsor central image reading; sponsor vigilance and literature screening (until check 11 is included); hospital sterile services audits.

First review · due after the first

3

C Checks in scope

Keep every check unless it truly does not apply. A check left out needs the reason, who agreed and what would bring it back. List the SRR risks whose controls each check verifies; an SRR control with no check here is a gap to raise.

#	Check	Included?	Control to verify (what must have happened)	Source record(s) used	SRR risk IDs covered	If not included: reason, agreed by, condition to include
1	**Participant entry: consent and eligibility** Consent signed on the current version before any study procedure; eligibility confirmed by the PI before surgery	☑ Included ☐ Not included ☐ Deferred until:	Signed consent on the version approved on the signing date; PI eligibility sign-off dated before surgery	Consent forms, PSEAR, EC approvals	R-01, R-02	
2	**Delegation and training** Everyone who did a study task was delegated and released for it on that date	☑ Included ☐ Not included ☐ Deferred until:	Each task in ICDL, PFVR and IQL done by someone delegated and released on that date	Delegation log, TRE, SCR	R-03, R-04, R-05	
3	**Sterile release to use** Integrity, expiry and storage confirmed before each device is used	☑ Included ☐ Not included ☐ Deferred until:	Expiry, package integrity and storage temperature recorded before opening	ICDL p. 1, accountability log, storage temperature log	R-06	
4	**Implant traceability and component reconciliation** Implanted components match labels, lot/UDI, accountability and CRF	☑ Included ☐ Not included ☐ Deferred until:	Every implanted component has REF and lot in the ICDL label sheet, accountability log and CRF	ICDL label sheet, accountability log, CRF implant page	R-07, R-08, R-17	
5	**Instrumentation and operating-room readiness** Instruments available, compatible and checked; substitutions recorded	☑ Included ☐ Not included ☐ Deferred until:	Loaner set checked complete the day before; any substitution recorded	ICDL readiness section, loaner-set check sheet	R-09	
6	**Operative documentation** Operative note lets a reader rebuild the final configuration and intraoperative decisions	☑ Included ☐ Not included ☐ Deferred until:	Note states final configuration, liner size and offset, and any deviation	Operative note, ICDL	R-10	
7	**Imaging usability and timing** Image sets acquired in window, with the required views, and usable	☑ Included ☐ Not included ☐ Deferred until:	Post-op and 3-month radiographs in window, required views, usable	IQL, PACS	R-11, R-12	

C Checks in scope (continued)

#	Check	Included?	Control to verify (what must have happened)	Source record(s) used	SRR risk IDs covered	If not included: reason, agreed by, condition to include
8	**Follow-up execution** Visits done in window with all endpoint-critical components	☑ Included ☐ Not included ☐ Deferred until:	Visits in window with Constant score and ASES complete	PFVR, FUCR	R-13	
9	**Endpoint data agreement** Endpoint-critical fields agree between source and CRF	☑ Included ☐ Not included ☐ Deferred until:	Constant score, ASES and implant position fields match the source	Source questionnaires, PFVR, CRF	R-14	
10	**Timeliness and closure** Notes, data entry, queries, safety follow-up and actions closed on time	☑ Included ☐ Not included ☐ Deferred until:	Operative note within 24 h, CRF entry within 5 working days, queries within 10 days, SAE within the CIP limit	EDC audit trail, query report, SART	R-15, R-16	
11	**External context** Safety notices, field actions and new literature on the device were looked for	☐ Included ☐ Not included ☑ Deferred until:		Sponsor safety letters, field notices	R-18	Sponsor screens literature and vigilance (CIP §12.4). Agreed by PI, 12 Aug 2026. Include at the first sponsor safety letter or any field notice on GLENOS.
12	**Complication monitoring** Complication reviews took place as planned and their signals were sent on	☑ Included ☐ Not included ☐ Deferred until:	CSST review held as planned and each signal sent on with a recipient ID	CSST, SCML	R-17, R-19	
13	Explant handling and return	☑ Included ☐ Not included ☐ Deferred until:	Explanted components labeled, quarantined and returned to the sponsor with an RMA number	SART, explant log, RMA forms	R-17	
14		☐ Included ☐ Not included ☐ Deferred until:				

Rows 13 and 14 are for a check the site adds because a risk in the SRR is not covered by checks 1–12. Use the same row number in C, D, E and F.

D Schedule, reviewer and sample

Time each check by how fast its failure would spread. The reviewer is named by role, released for the task, and does not review work they did. Unit and volume: what is counted (procedure, image set, visit, field, event…) and how many are expected per period. Selection method: how the units reviewed are chosen. Sample size: the minimum reviewed each cycle, and when every unit is reviewed instead (a small population is reviewed in full). Expansion trigger: what makes the sample larger.

#	Check	Timing basis	Routine frequency	Early-review trigger	Reviewer / backup (roles)	Unit and volume	Selection method	Sample size	Expansion trigger
1	**Participant entry: consent and eligibility** Consent signed on the current version before any study procedure; eligibility confirmed by the PI before surgery	☐ Calendar ☑ Case count ☐ Each event ☐ Milestone	Each of the first 5 inclusions, then quarterly	Re-consent due after an amendment	Research nurse (P-04) / PI	Participant; about 4 per quarter	All new inclusions	All (fewer than 10)	Any version mismatch: all participants
2	**Delegation and training** Everyone who did a study task was delegated and released for it on that date	☑ Calendar ☐ Case count ☐ Each event ☐ Milestone	Quarterly	New staff member or new task delegated	Sub-I (P-02) / PI	Study task; about 60 per quarter	Every 5th task in the task list, random start	10 tasks	A task done before release: all tasks of that person
3	**Sterile release to use** Integrity, expiry and storage confirmed before each device is used	☐ Calendar ☐ Case count ☑ Each event ☐ Milestone	Every procedure	Temperature excursion or damaged package	OR nurse lead (P-08) / coordinator	Procedure; about 4 per quarter	All	All	
4	**Implant traceability and component reconciliation** Implanted components match labels, lot/UDI, accountability and CRF	☐ Calendar ☐ Case count ☑ Each event ☐ Milestone	Within 5 days of each procedure	Lot or REF missing in any record	Coordinator (P-06) / research nurse	Implanted component; about 16 per quarter	All	All	
5	**Instrumentation and operating-room readiness** Instruments available, compatible and checked; substitutions recorded	☐ Calendar ☑ Case count ☐ Each event ☐ Milestone	First 3 procedures per surgeon, then every 5th	Instrument malfunction or substitution	OR nurse lead / Sub-I	Procedure	Every 5th procedure by date	At least 1 per quarter	Unrecorded substitution: all since last review
6	**Operative documentation** Operative note lets a reader rebuild the final configuration and intraoperative decisions	☐ Calendar ☑ Case count ☐ Each event ☐ Milestone	First 3 per surgeon, then quarterly	A new surgeon starts	Sub-I / PI	Operative note; about 4 per quarter	Latest note of each surgeon	2 per quarter	Liner size or offset missing: all
7	**Imaging usability and timing** Image sets acquired in window, with the required views, and usable	☑ Calendar ☐ Case count ☐ Each event ☐ Milestone	Monthly	More than 1 in 5 image sets repeated	Radiology research tech (P-09) / research nurse	Image set; about 8 per month	All	All (fewer than 10)	

D Schedule, reviewer and sample (continued)

#	Check	Timing basis	Routine frequency	Early-review trigger	Reviewer / backup (roles)	Unit and volume	Selection method	Sample size	Expansion trigger
8	**Follow-up execution** Visits done in window with all endpoint-critical components	☑ Calendar ☐ Case count ☐ Each event ☐ Milestone	Quarterly	Missed visit	Research nurse / coordinator	Visit; about 12 per quarter	Random 6 from the PFVR visit list	6; all if fewer than 10	A missed endpoint component: all
9	**Endpoint data agreement** Endpoint-critical fields agree between source and CRF	☑ Calendar ☐ Case count ☐ Each event ☐ Milestone	Quarterly	Discrepancy in an endpoint field	Research nurse (P-04) / Sub-I	Endpoint field; about 120 per quarter	All fields of 5 visits picked at random	5 visits; all affected fields after a discrepancy	2 discrepancies in one review
10	**Timeliness and closure** Notes, data entry, queries, safety follow-up and actions closed on time	☑ Calendar ☐ Case count ☐ Each event ☐ Milestone	Monthly	An SAE reported late	Coordinator / PI	Item with a deadline	All SAE reports; queries open over 10 days	All SAE; 10 queries	One late SAE report: every deadline that month
11	**External context** Safety notices, field actions and new literature on the device were looked for	☐ Calendar ☐ Case count ☐ Each event ☑ Milestone	From the first sponsor safety letter	Field notice on GLENOS	PI / Sub-I	Search			
12	**Complication monitoring** Complication reviews took place as planned and their signals were sent on	☑ Calendar ☐ Case count ☐ Each event ☐ Milestone	Quarterly; monthly for 3 months after a CSST signal	SADE, or 2 dislocations in a quarter	Sub-I / PI	CSST review	All	All	
13	Explant handling and return	☐ Calendar ☐ Case count ☑ Each event ☐ Milestone	Each explant	Explant not returned within 30 days	Coordinator / OR nurse lead	Explanted component	All	All	
14		☐ Calendar ☐ Case count ☐ Each event ☐ Milestone							

Rows 13 and 14 are for a check the site adds because a risk in the SRR is not covered by checks 1–12. Use the same row number in C, D, E and F.

E Criteria the reviewer applies

Write each criterion so two reviewers would reach the same answer. The IMC rates each check on these three questions separately.

#	Check	Execution: the control was done when…	Evidence: the record is sufficient when…	Effective: the control worked when…	Records the reviewer cites in the IMC
1	**Participant entry: consent and eligibility** Consent signed on the current version before any study procedure; eligibility confirmed by the PI before surgery	Consent and PI sign-off exist for every participant reviewed	Signed, dated pages present; version legible	No procedure before consent; no surgery before PI sign-off	PSEAR IDs, consent version
2	**Delegation and training** Everyone who did a study task was delegated and released for it on that date	Each sampled task matches a delegated, released person	Initials traceable in the delegation log	No task done before the release date	Delegation log rev., TRE IDs
3	**Sterile release to use** Integrity, expiry and storage confirmed before each device is used	Checks recorded before opening for every device	ICDL p. 1 signed; temperature log for the day	No expired or breached device used	ICDL IDs
4	**Implant traceability and component reconciliation** Implanted components match labels, lot/UDI, accountability and CRF	REF and lot in all three records	Label sheet attached to the ICDL	Records agree; implanted matches planned or the change is explained	ICDL IDs, CRF pages
5	**Instrumentation and operating-room readiness** Instruments available, compatible and checked; substitutions recorded	Set check done the day before	Signed check sheet	No delay or substitution from a missing instrument	ICDL IDs
6	**Operative documentation** Operative note lets a reader rebuild the final configuration and intraoperative decisions	Note signed within 24 h	Configuration, sizes and deviations stated	A reader can rebuild the procedure from the note alone	Note dates, ICDL IDs
7	**Imaging usability and timing** Image sets acquired in window, with the required views, and usable	Images acquired in window	IQL entry with views and quality	Usable for the endpoint without repeat	IQL IDs

E Criteria the reviewer applies (continued)

#	Check	Execution: the control was done when…	Evidence: the record is sufficient when…	Effective: the control worked when…	Records the reviewer cites in the IMC
8	**Follow-up execution** Visits done in window with all endpoint-critical components	Visit done in window	PFVR complete, endpoint components marked	No endpoint component missing	PFVR IDs
9	**Endpoint data agreement** Endpoint-critical fields agree between source and CRF	Fields compared for each sampled visit	Source questionnaire kept	No discrepancy, or each corrected and explained	PFVR IDs, CRF pages
10	**Timeliness and closure** Notes, data entry, queries, safety follow-up and actions closed on time	Dates retrieved for each item	Audit trail export kept	All within the limits in C	Report export date
11	**External context** Safety notices, field actions and new literature on the device were looked for				
12	**Complication monitoring** Complication reviews took place as planned and their signals were sent on	Review held in the planned month	Signed CSST with snapshot ID	Each signal has a recipient ID	CSST IDs
13	Explant handling and return	Explant labeled and quarantined on the day	Explant log and RMA form	Returned within 30 days with lot readable	SART IDs, RMA numbers
14					

Rows 13 and 14 are for a check the site adds because a risk in the SRR is not covered by checks 1–12. Use the same row number in C, D, E and F.

F Triggers for an earlier review

Any trigger marked Yes calls a review before the next scheduled one, for the checks listed. A minor observation is not a trigger.

Trigger	Applies?	Checks reviewed early (#)	Called by (role)	Within (days)
Serious adverse event or serious device effect at the site	☑ Yes ☐ No	3, 4, 6, 10, 12, 13	PI	5
Complication cluster or signal from the CSST	☑ Yes ☐ No	5, 6, 12	PI	10
Important protocol deviation	☑ Yes ☐ No	1, 2 and the related check	Coordinator	10
Repeated endpoint data discrepancy	☑ Yes ☐ No	8, 9	Research nurse	10
Pattern of unusable or late images (IQL)	☑ Yes ☐ No	7	Radiology tech	14
Device, instrument or equipment malfunction	☑ Yes ☐ No	3, 5	OR nurse lead	5
CIP amendment	☑ Yes ☐ No	1, 2, 8	Coordinator	30
Departure or change of a key person or of a reviewer	☑ Yes ☐ No	2	PI	14
Sponsor finding that needs a local response	☑ Yes ☐ No	As named in the finding	PI	14
Corrective action that failed or was reopened	☑ Yes ☐ No	The check that raised it	PI	14
Explant not returned to the sponsor within 30 days	☑ Yes ☐ No	4, 13	Coordinator	5

G Where each result goes

For each result the IMC can give, write the minimum response the site commits to, the record it goes to, and who owns it.

Result in the IMC	Minimum response and record it goes to	Owner (role)
Complete, adequate and effective	None; the result stays in the IMC.	Reviewer
Observation: isolated, low consequence, corrected at review	Fixed at the review; IMC action log with date.	Reviewer
Partial execution or partial evidence	Corrected in the owning record (ICDL, PFVR, IQL...); rechecked at the next review.	Owner of the record
Not performed, or evidence missing	IMC action log; coordinator decides within 5 days whether it is a deviation to report.	Coordinator
Not effective	CAPA log; SRR rating revisited at the next SRR review.	PI
Not assessable	Repeated within 30 days, or with a larger sample.	Reviewer
Recurrent, across units, or with material consequence	CAPA log; PI decision recorded in the POT; SRR updated.	PI
Possible harm to a participant or a reportable event	PI and SART the same day; sponsor as the CIP requires; IMC keeps only the reference.	PI / coordinator

H Ready to start

Confirm before the first inclusion. PI approval means the plan can be run with the people and time available.

Confirmation	Status	Date, or action and due date if No
Every reviewer and backup is released for the task in the delegation log	☑ Yes ☐ No	Released 30 Oct 2026 (TRE-014-11 for P-09)
Staff who do the checked work know the plan	☐ Yes ☑ No	OR nurses not yet briefed on check 13: briefing 11 Nov 2026 (M. Silva)
The IMC is prepared with this IMP ID and version	☑ Yes ☐ No	IMC-014-03 set up on 3 Nov 2026
First review due (date or after inclusion number)	IMC-014-03 due 30 Nov 2026 (first review under v1.1); v1.0 first review IMC-014-01 held 30 Sep 2026 after inclusion 3	

Plan owner

Name / role: Maria Silva / Study Coordinator

Signature / date:

M. Silva 3 Nov 2026

Principal Investigator

Name: Dr Alex Morgan

Signature / date:

A. Morgan 3 Nov 2026

I Changes to the plan

A change applies from its effective date. A result already seen is never the reason to loosen a rule for the same period.

Change ID	Section / check	Previous rule	New rule	Reason (IMC, CAPA, SRR, sponsor…)	Effective from	Approved by
CH-01	D · 9	5 visits per quarter, no expansion	5 visits; all affected fields after a discrepancy; expand at 2	IMC-014-02: 3 ASES discrepancies; CAPA-014-003	4 Nov 2026	A. Morgan
CH-02	C, D, E · 13; F · 11	No check on explants	Check 13 added: explant handling and return	SART-014-005: explant lot missing in the ICDL; SRR R-17	4 Nov 2026	A. Morgan
CH-03	D · 12	Quarterly	Monthly for 3 months after any CSST signal	CSST-014-02 dislocation signal	4 Nov 2026	A. Morgan
CH-04	D · 2	Reviewer: coordinator	Reviewer: Sub-I; the coordinator keeps the log	IMC-014-02: coordinator reviewed her own delegation entries	4 Nov 2026	A. Morgan